

Introducion

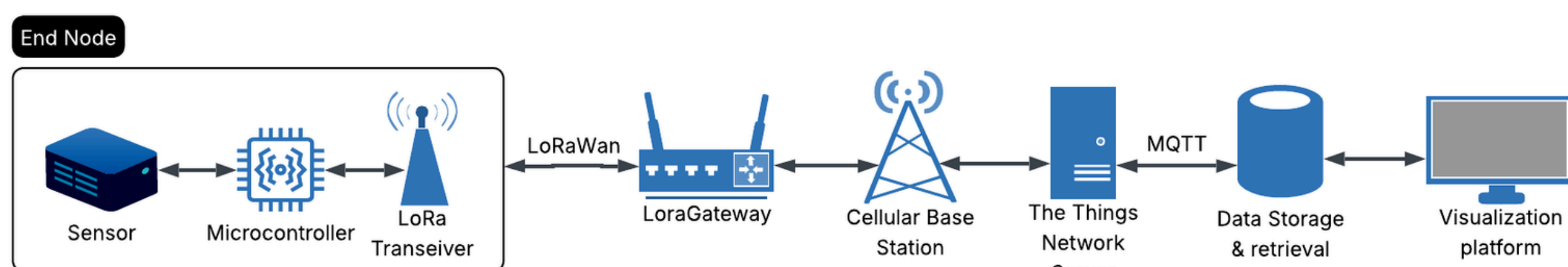
River water level data is critical for flood warning and water resource management. But the rivers that need monitoring most are often in remote areas, without cellular coverage, and require a low-cost system that can run independently on limited battery power. This project builds a complete monitoring system using LoRaWAN for long-range, low- power data transmission, comparing three non-contact sensors (Ultrasonic, Radar, and LiDAR) for accuracy, performance, limitations, and overall reliability in remote river settings.

Objectives

- Evaluate and compare the performance of Ultrasonic, Radar, and LiDAR sensors under various field conditions.
- Implement a robust communication architecture to transmit real-time data wirelessly from remote areas.
- Optimize power consumption and extend battery life using a dynamic data transmission strategy.

Methodology

System Architecture



FFig. 1: System architecture, from the sensor node to the visualization platform via LoRaWAN.

Adaptive Transmission Strategy

The system dynamically adjusts the transmission interval based on river activity, maximizing the sleep interval when the water level is stable and increasing frequency during rapid changes, reducing power consumption and cost without missing critical flood data.

Condition	Change Between Readings	Transmission Interval
Stable flow	<1% of distance	15 minutes
Moderate activity	1%–3.5% of distance	5 minutes
Rapid change	>3.5% of distance	1 minute

Table 1: Adaptive transmission intervals based on river activity level.

Sensors

All three sensors share the same working principle, each emits a pulse and measures how long it takes to reflect back from the water surface.

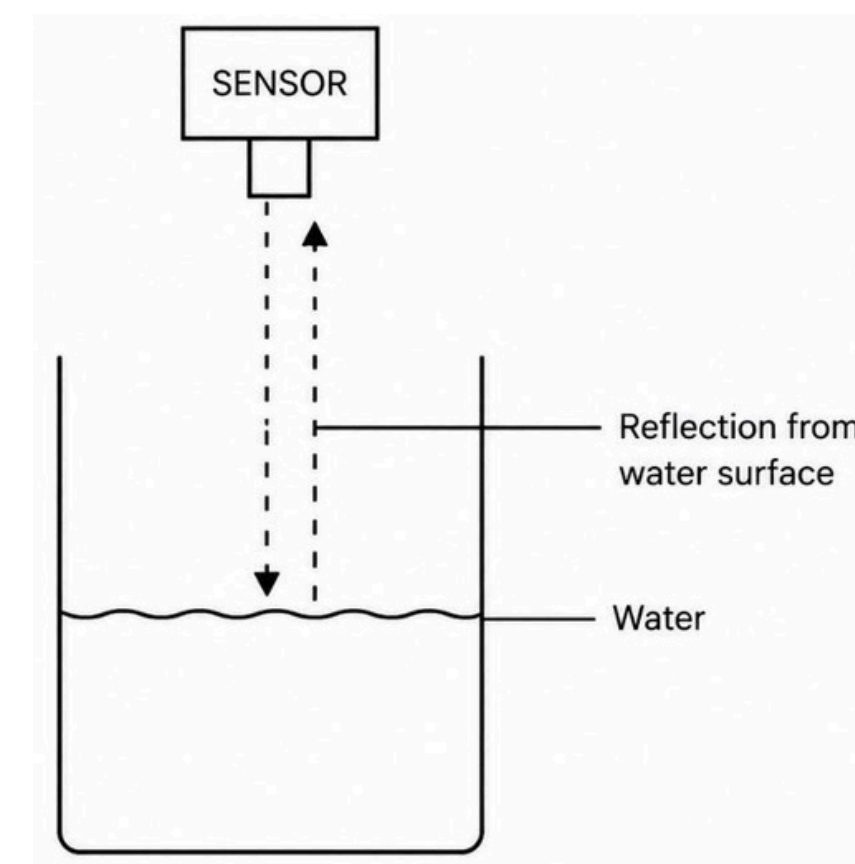


Fig. 2: Time-of-flight measurement principle

Timavo River Experiment

To evaluate sensor performance, a field experiment was conducted at the Timavo River, located in the Province of Trieste, Italy.



Fig. 3: Field setup at the Timavo River, Trieste, Italy, with all three sensors mounted side by side.

Sensor	Model	Wave Type	Range	Beam Width
Ultrasonic	Milesight EM500-UDL	Sound	0.25–10 m	~60°
Radar	Milesight EM410-RDL	Electromagnetic	0.3–12 m	8°
LiDAR	Garmin LiDAR-Lite v3HP	Electromagnetic	up to 40 m	~0.5°

Table 2: Specifications of the tested sensors.

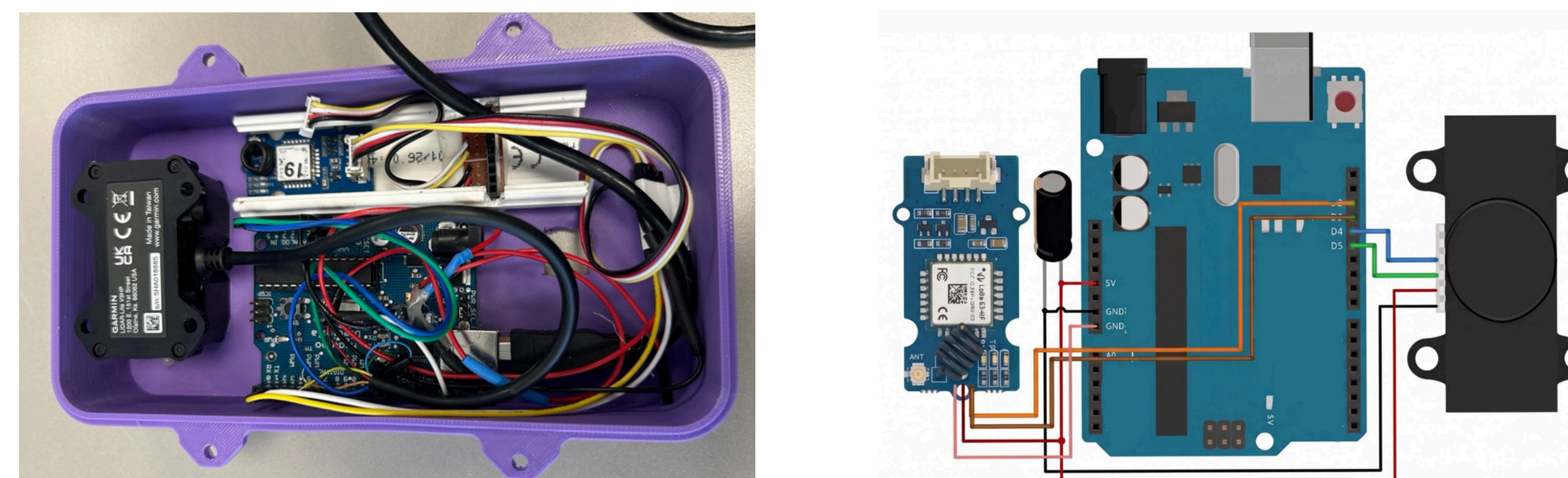


Fig. 4: Custom-built LiDAR node

Results

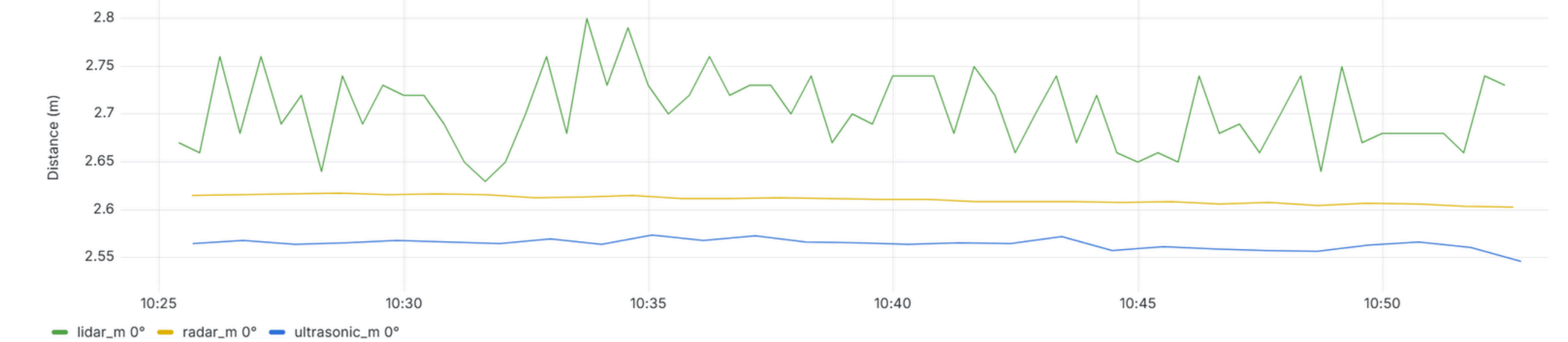


Fig. 5: Sensors Performance and Measurement Variations Over Time

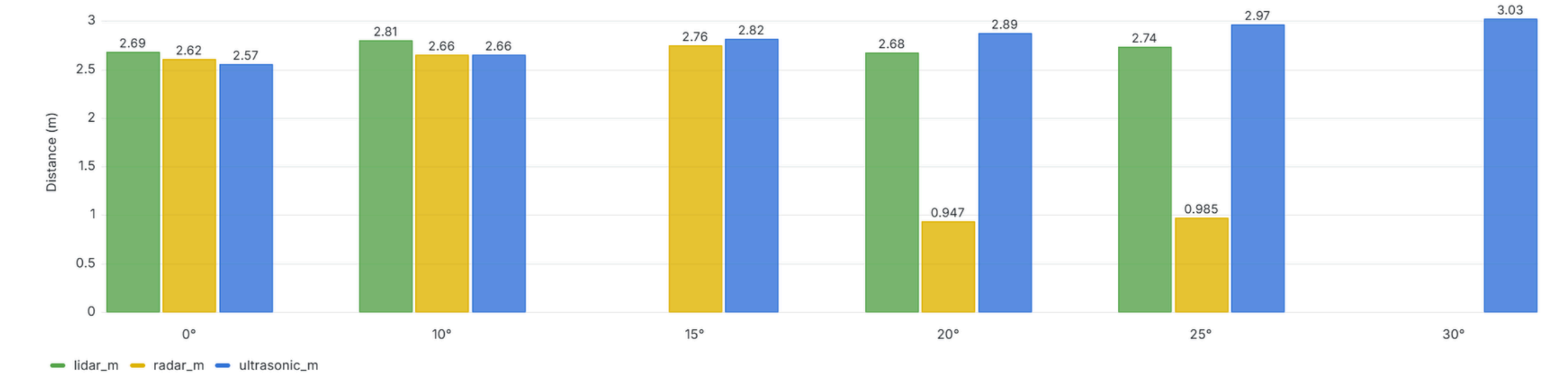


Fig. 6: Average Distance Measured by Each Sensor at Different Tilt Angles

- Radar : Delivered the most stable and repeatable readings when aligned within its 15° rated tilt range. Beyond this threshold, accuracy degraded sharply (~66.5% error).
- Ultrasonic : Maintained high accuracy across the full tested tilt range (up to 30°); however, its wide beam increases sensitivity to false echoes from nearby structures, requiring a clear, obstruction-free installation zone.
- LiDAR : Achieved the longest effective range, with accuracy improving in turbid water. In clear water, the narrow beam tends to penetrate the surface rather than reflect back, also making it more sensitive to surface changes and resulting in the highest measurement variation among the three sensors.

Future work

- Sensors were not tested under turbulent flow. Next step: repeat the trial during higher-flow or post-rainfall conditions, when accurate readings matter most for flood warning.
- Battery and solar performance under the adaptive scheme were not evaluated. Next step: long-term field testing to assess power sustainability and reliability.

Conclusion

Three non-contact sensors were evaluated and integrated into a complete LoRaWAN-based monitoring system with adaptive data transmission. Radar provided the highest precision, ultrasonic offered the greatest installation tolerance, and LiDAR achieved the longest range. These findings offer practical guidance for sensor selection and confirm that reliable, low-power flood monitoring is achievable in remote areas without cellular infrastructure.

References

- J. D. Paul, W. Buytaert, N. Sah, "A Technical Evaluation of Lidar-Based Measurement of River Water Levels," Water Resources Research 56.4 (2020), e2019WR026810.
- J. N. Kabi et al., "Low cost, LoRa based river water level data acquisition system," HardwareX 14 (2023), e00414.